

Kidanemaryam Wagaw Reta

Plant Physiologist | Plant Biochemist | Predictive Modeling | System Crop Scientist | Multi-Omics Integration

Ben-Gurion University of the Negev, Be'er Sheva, Israel

kidane.et@gmail.com +972587964242 [Google Scholar](#) [Researchgate](#) [ORCiD](#) [Linkedin](#) [X/Twitter](#)

Brief summary

Plant scientist with a PhD in plant biotechnology, specializing in crop physiology, plant metabolism, and trait characterization under complex abiotic stresses. My PhD research focused on how root system variation drives whole-plant performance, integrating ion partitioning, photosynthesis, chlorophyll fluorescence, metabolism, gene expression, and phenology to explain and predict stress resilience and yield outcomes. My work specifically examines how photosynthetic performance, water status, and metabolic regulation interact to determine crop productivity under climate-driven stress.

I have designed and executed multi-season experiments across diverse field conditions, controlled greenhouse and laboratory systems, and semi-controlled environments (net house and phenotyping platforms), quantifying physiological, metabolic, molecular, and agronomic responses under complex stress interactions. My work combines high-resolution gas exchange and chlorophyll fluorescence measurements (LI-COR, MicroPAM) with metabolomics (GC-MS/LC-MS), ionomics (ICP-OES), and advanced data analysis, including network modeling and machine learning (R, Python), to identify key traits and define performance thresholds. I have applied these approaches across diverse crop systems, including grapevine and major cereals (sorghum, maize, barley, wheat, beans), contributing to large-scale germplasm characterization and the development and release of improved sorghum varieties for dryland production. I have additional expertise in seed biology, including quantitative germination, dormancy, and vigor assays under environmental and mechanical constraints, linking seed performance with physiological resilience and crop establishment. My research also integrates root phenotyping and root-shoot trait analysis with physiological and metabolic datasets to link belowground processes with whole-plant performance under abiotic stress.

My research work also includes defining stress-response thresholds and tipping points through integrative physiology, metabolomics, phenology, agronomic performance, and network analysis to identify key traits and define physiological, metabolic, and agronomic stress thresholds associated with resilience and yield performance.

Currently, I lead an externally funded research project integrating multi-year physiological phenotyping, UAV-based thermal sensing, metabolomics, and gene expression profiling to characterize varietal responses to heatwave stress and identify resilient genotypes through statistical, bioinformatics, and network-based analyses. As part of this effort, I initiated an interdisciplinary collaboration with computer scientists to develop AI-enabled predictive models that integrate biological datasets with climate forecasting data to dynamically predict long-term varietal performance under future environmental scenarios.

Alongside my research, I have led and coordinated multi-institutional, internationally funded research projects and activities, and mentored students, technicians, and junior researchers, supporting experimental design, data analysis, and scientific development within collaborative, multidisciplinary teams. I bring a strong experimental and analytical approach to building and optimizing trait-screening strategies, while contributing to collaborative team environments that deliver robust, reproducible results for crop improvement.

I am particularly interested in scaling physiological and metabolic responses from the cellular to whole-plant and field levels using data-driven approaches, including machine learning and high-throughput phenotyping.

Career trajectory

My research trajectory has evolved from agronomy, seed biology, and quantitative breeding to mechanistic plant physiology and multi-omics-driven systems biology. This progression has built an integrated expertise spanning field and controlled-environment experimentation, trait discovery, metabolomics, molecular analysis, predictive modeling, network analysis, and machine learning to advance understanding of crop resilience under abiotic stress and to develop improved crop varieties that can adapt to the growing impacts of climate change.

Keywords: Plant Physiology, Crop Stress Biology, Plant Metabolomics, Root-Shoot Interactions & Phenotyping, Demand Led Breeding, Phenotyping, Crop Improvement and Trait Discovery, Multi-Omics Integration, Root-Shoot and Stress Interactions, Predictive Modeling and Machine Learning, Network Analysis, G x E Interactions, Stress Thresholds and Crop Resilience, Seed Biology, Climate-informed AI predictive modeling

Core Expertise

Curriculum Vitae

- ✓ Crop physiology: Photosynthesis & gas exchange (LI-COR), chlorophyll fluorescence (MicroPAM), above- and below-ground trait characterization
- ✓ Metabolite profiling, identification, spectral curation, and analytical method development (GC-MS/LC-MS, ICP-OES, HS-SPME-GC-MS)
- ✓ Stress threshold identification: defining physiological, metabolic, phenology, and agronomic stress thresholds, tipping points, and trait-performance breakpoints
- ✓ Root phenotyping and root system architecture analysis linked to whole-plant physiology, ion regulation, and stress resilience
- ✓ AI-enabled predictive modeling and digital phenotyping for climate-resilient crop performance
- ✓ Phenotyping pipeline development and optimization
- ✓ Seed physiology: germination, dormancy, vigor assessment under environmental and mechanical constraints
- ✓ Predictive modeling and bioinformatics (R, Python)
- ✓ Stress threshold identification, tipping-point analysis, and network-based trait discovery
- ✓ Metabolic network inference and pathway analysis
- ✓ Root physiology and rootstock-mediated trait modulation in non-model crop species
- ✓ Trait screening and validation across a large germplasm panel (sorghum, grapevine, maize, barley, wheat, beans)
- ✓ Multi-omics integration (metabolomics, ionomics, transcriptomics, phenomics)
- ✓ Crop improvement: trait validation, selection, and lab-to-field translation

Research interests

- ✓ Crop physiology and biochemical mechanisms governing trait performance under abiotic stress
- ✓ Root system function, ion selectivity, and rhizosphere dynamics for trait development
- ✓ Carbon-nitrogen interactions and metabolic regulation of photosynthesis under stress
- ✓ Scaling physiological traits from leaf to canopy and field performance
- ✓ Soil-plant interaction under multifactorial stress combination-linking soil analysis to plant performance
- ✓ Integration of high-throughput phenotyping (UAV, hyperspectral) with physiological traits
- ✓ Digital agriculture, AI-assisted crop forecasting, and climate-resilience modeling
- ✓ Gas exchange and photosynthetic performance under abiotic stress: mechanisms and trait thresholds
- ✓ Integration of physiological, biochemical, and omics data to validate and improve crop traits across germplasm
- ✓ Predictive trait-performance relationships, threshold behavior, and network regulation across diverse germplasm and environments
- ✓ Root system architecture and plant–soil interactions in stress adaptation
- ✓ Lab-to-field translation of mechanistic insights for scalable crop improvement and commercial deployment

Education

PhD in Plant Metabolomics, Ben-Gurion University of the Negev (Oct 2021 – Nov 2025), Israel.

Thesis title: Responses of *Vitis vinifera* grafts to combined abiotic stress: Physiology and Metabolism.

- ✓ Designed and executed multi-season experiments across field, greenhouse, and semi-controlled systems to quantify crop responses to combined salinity, heatwave, and water stress
- ✓ Performed metabolite identification, spectral curation, annotation, quantification, and integrative metabolomics analyses (GC-MS/LC-MS, HS-SPME-GC-MS) to characterize stress-induced metabolic reprogramming and trait-associated biochemical signatures.
- ✓ Quantified photosynthetic performance and stress responses using gas exchange and chlorophyll fluorescence (LI-COR, MicroPAM), resolving genotype-specific physiological dynamics
- ✓ Integrated physiology with multi-omics (metabolomics, transcriptomics, ionomics, phenomics) to identify mechanisms and traits associated with stress resilience
- ✓ Integrated root-related traits with physiological, ionomic, and metabolomic analyses to resolve how belowground processes influence stress adaptation and crop performance
- ✓ Explored links between carbon metabolism and ion regulation, providing a foundation for extending analysis toward carbon-nitrogen interactions under stress
- ✓ Developed predictive models linking environmental variables, physiological traits, and molecular responses to crop performance
- ✓ Applied advanced statistical and systems approaches to define physiological and agronomic stress thresholds, revealing critical transition points associated with performance decline and stress adaptation
- ✓ Performed metabolic network and pathway analysis to identify key regulatory nodes underlying stress adaptation
- ✓ Identified and validated physiological and metabolic traits associated with stress tolerance across grafted genotypes

Curriculum Vitae

- ✓ Contributed to transcriptomic–metabolomic integration for functional characterization of genes involved in central carbon metabolism

MSc in Plant Breeding, Jimma University (Oct 2016 – Oct 2018), Ethiopia

Thesis title: Assessment of heterotic performance and combining ability of Ethiopian elite sorghum (*Sorghum bicolor* (L.) Moench) lines.

- ✓ Designed and conducted field-based experiments for hybrid development in sorghum, evaluating heterosis and combining ability across elite germplasm
- ✓ Applied quantitative genetics and multi-environment trial analysis to assess trait performance, stability, and genotype × environment interactions
- ✓ Identified high-performing parental lines and hybrids to support selection and advancement within breeding programs

BSc in Plant Sciences (Great Distinction), Haramaya University (Sept 2009 – Jul 2012), Ethiopia.

Experience

Postdoctoral researcher in Plant Metabolomics and Physiology, Ben Gurion University of the Negev (Nov 2025 – Present), Be'er Sheva, Israel.

- ✓ Lead an OIV-funded integrative phenotyping project combining UAV-based thermal imaging, physiology, metabolomics, gene expression, and AI-assisted predictive modeling to identify climate-resilient varietal performance and develop dynamic forecasting tools for stress adaptation
- ✓ Design and execute above- and below-ground crop physiology experiments (gas exchange, chlorophyll fluorescence, ionomics, plant biochemistry) to quantify genotype- and stage-dependent trait variation across controlled and field environments
- ✓ Led advanced metabolite annotation, spectral curation, quantification, and systems-level metabolomics analyses, integrating biochemical, physiological, and multi-omics datasets to identify regulatory pathways and predictive signatures of stress resilience
- ✓ Integrate multi-omics datasets (metabolomics, transcriptomics, ionomics, phenomics) to resolve carbon metabolism, ion partitioning, and biochemical regulation underlying stress responses in non-model crop systems
- ✓ Generate and analyze metabolomics and volatilomics datasets (GC-MS, LC-MS, HS-SPME-GC-MS) using multivariate statistics and machine learning approaches
- ✓ Integrated root and shoot phenotyping with multi-omics and environmental datasets to identify trait interactions underlying crop resilience across scales
- ✓ Identify and validate physiological, metabolic, and phenological traits and thresholds linking root-zone processes, environmental stress, and measurable crop performance outcomes
- ✓ Developed threshold-based and network-informed approaches to define physiological and agronomic stress thresholds and identify critical tipping points governing crop resilience under abiotic stress
- ✓ Apply network and systems-level analysis of central and specialized metabolism to infer regulatory relationships and context-dependent metabolic interactions
- ✓ Integrate UAV-based thermal imaging with physiological measurements to link canopy dynamics with cellular and biochemical responses under stress
- ✓ Conduct multi-year field and controlled-environment experiments to characterize crop performance across phenological stages under abiotic stress conditions
- ✓ Conducted metabolite identification, spectral curation, profiling, quantification, annotation, and analysis

Researcher II (Plant Molecular Breeder), Bio and Emerging Technology Institute (Feb 2021 – to date), Addis Ababa, Ethiopia.

- ✓ Led crop improvement research for dryland production systems, integrating physiological phenotyping, quantitative genetics, and molecular tools to identify and validate stress-resilient performance traits across diverse germplasm and environments
- ✓ Applied genomic-assisted selection and molecular breeding approaches to accelerate trait validation, selection, and deployment of improved crop varieties adapted to abiotic stress conditions
- ✓ Coordinated multi-institutional, internationally funded research programs (BIOINNOVATE Africa), contributing to experimental design, project implementation, scientific reporting, and dissemination of research outputs

Junior Researcher – Associate Researcher (Plant Breeder), Ethiopian Institute of Agricultural Research (May 2014 – January 2021), Addis Ababa, Ethiopia.

Curriculum Vitae

- ✓ Designed and implemented breeding schemes, including crossing, hybridization, and heterosis exploitation, leading to the development and release of three improved sorghum varieties, and contributed to 8 more released varieties as a team.
- ✓ Led large-scale phenotyping and physiological characterization of sorghum germplasm, linking transpiration efficiency, root architecture, stress responses, and agronomic traits to yield performance across multi-environment trials
- ✓ Developed and analyzed multi-environment datasets to resolve genotype $\times$ environment interactions and identify stable, high-performing genotypes under abiotic stress conditions
- ✓ Integrated field phenotyping with SNP-based genotyping (GBS) and genotype-phenotype association analysis to support trait dissection in sorghum breeding populations
- ✓ Contributed to data-driven selection pipelines combining statistical modeling and machine-learning approaches for trait selection and performance prediction

Crop production and protection expert (System Agronomist), Amhara Bureau of Agriculture, Amhara Sayint Agricultural Office (October 2012 – April 2014), Amhara Sayint, Ethiopia.

- ✓ Provided field-level agronomic diagnostics and advisory support for major staple crops (maize, wheat, barley, and haricot bean), addressing yield constraints under smallholder farming systems
- ✓ Conducted seed quality assessment and certification for haricot bean production systems targeting both local and export markets, supporting varietal quality control and deployment
- ✓ Implemented crop production and protection programs across multiple crop species, contributing to integrated pest and stress management at the field level
- ✓ Seed certification for the export market and dispatching to the port warehouse

Scientific publications (Publications may appear under the names: Reta K., Wagaw K., and Wagaw R.K., depending on journal format)

Selected publications (for the full list of publications, [click here](#))

- Reta K.**, Lupo Y., Persi N.S., Lazarovitch N. and Fait A. (2025). Modulation of phenology and agronomical performance of Syrah grafted on two rootstocks under combined salinity and water stress conditions: A three-year field study. *Plant Stress*, Elsevier B.V., Vol. 18, p. 101050, doi: 10.1016/j.stress.2025.101050 (IF=6.9).
- Reta K.**, Lazarovitch N. and Fait A. (2025). Integrated metabolomics, ionomics, physiology, and network analysis reveals distinct salinity tipping point in *Vitis vinifera* cv. Syrah to enter a stress response mode. *Plant Stress*, Elsevier B.V., Vol. 16, pp. 1–14, doi: 10.1016/j.stress.2025.100864 (IF=6.9).
- Reta K.**, Netzer Y., Lazarovitch N. and Fait A. (2025). Canopy management practices in warm environment vineyards to improve grape yield and quality in a changing climate. A review A vademecum to vine canopy management under the challenge of global warming. *Scientia Horticulturae*, Elsevier B.V., Vol. 341, p. 113998, doi: 10.1016/j.scienta.2025.113998 (IF=4.2).
- Zohar Y., **Reta K.**, Drori E., Gliksman U., Rauchberger S., Bar E., et al. (2024). Improved berry and wine quality of *Vitis vinifera* L. cv. Gewürztraminer grown in an arid climate using a Y-shaped training system. *OENO One*, Vol. 58 No. 1, pp. 1–15, doi: 10.20870/oeno-one.2024.58.1.7148 (IF=2.2).
- Wagaw K.** (2019). Review on Mechanisms of Drought Tolerance in Sorghum (*Sorghum bicolor* (L.) Moench) Basis and Breeding Methods. *Acad. Res. J. Agri. Sci. Res*, Vol. 7 No. 2, pp. 87–99, doi: 10.14662/ARJASR2019.007 (IF = 8.4).

Papers pending publication

- Reta K.**, Lupo Y., Sikron-Persi N., Acuna T., Lazarovitch N., Fait A. Contrasting metabolic and physiological responses underlie rootstock-mediated tolerance to combined salinity and water-deficit stress in cv.Syrah. (Under review in *Plant Stress*, IF=6.9).
- Reta K.**, Caras T., Sikron-Persi N., Bustan A., Lazarovitch N., Paz-Kagan T., Fait A. Integrative phenotyping, combining metabolomics, molecular biology, and remote sensing, revealed a varietal signature in response to a heatwave event in arid viticulture. (In prep for submission to *Plant Phenomics*, IF=6.3).
- Reta K.**, Verma P., Gashu K., Bustan A., Fait A. Identification of high temperature-induced coordinated metabolic and phenological shifts in a varietal collection of wine grapes. (Under preparation for *PCE*, IF=6.3).
- Reta K.**, Lazarovitch N., and Fait A. Comparative machine learning reveals rootstock-modulated metabolic and temporal signatures in grafted grapevine under combined salinity and water deficit stress. (Under preparation for *Horticulture Research*, IF=8.5).
- Degu A., Faigenboim A., **Reta K.**, Dynkin I., Fait A., and Reshef N. Functional Characterization of Grapevine PEPC and Cytosolic MDH Genes Reveals Divergent Regulation of Malate and Sugar Metabolism (Under review in

Curriculum Vitae

Horticulture Research, IF =8.5)

Javali A., **Reta K.**, Herzberg M., and Fait A. Engineered multi-ionic irrigation reveals cultivar-specific coordination of root architecture, ion partitioning, and metabolism in grapevine. (In preparation)

Research and working manuals

Seyoum A., Nega A., Tadesse T., Zerihun M., **Wagaw K.**, Gebreyohannes A., et al., 2019 Sorghum Breeding Manual. Ethiopian Institute of Agricultural Research, ISBN 978999466658

Projects

- ✓ **PI** - Integrative metabolomics, physiology, and predictive modeling in *Vitis vinifera* (L.) under elevated temperature: From physiology to metabolic and UAV-based thermal phenotyping (Funded by OIV, 2023-2026).
- ✓ **Coordinator** - Small-scale community biorefining of sorghum for food, sugar, and biomaterials in rainfed areas of Eastern Africa (Funded by BIOINOVATE Africa, 2019-2022).
- ✓ **Co-PI** - Climate-smart interventions for smallholder farmers in Ethiopia (CultiAF-2) in collaboration with the University of Queensland, Australia, and Regional Research institutes (Funded by IDRC and ACIAR, 2019-2021).
- ✓ **Component 2 focal person** - Modernizing Ethiopian Agricultural Research in Crop Improvement (MERCI) in collaboration with the University of Queensland, Australia, and Regional Research institutes (Funded by BMGF, 2016-2020).

Scientific presentations at international conferences

Selected oral presentations (for the full list of scientific presentations, oral & poster, [click here](#))

- ✓ The rebirth of heroic viticulture in Israel. Oral: The 52nd Annual Ben-Gurion University (BGU) Board of Governors meeting (18/05/2022), Ben-Gurion University, Israel.
- ✓ Rootstock-Driven Metabolic, Physiological and Agronomical Responses Determine Scion Tolerance under Combined Salinity and Water-Deficit Stress: A Long-Term Field Experiment. Oral; DDD: Drylands, Deserts and Desertification 2025, (Nov 30-Dec 04/2025), Ben Gurion University of the Negev, Israel.
- ✓ Rootstock-mediated responses of grapevine *Vitis vinifera* (L.) metabolism and physiology to combined water deficit and salinity stress in Syrah grafts. Oral; ENO Macrowine 2023, (10-13/July/2023), Bordeaux, France.
- ✓ Combined abiotic stress in cv.Syrah grafted onto different rootstocks: a Metabolic and physiological response. Oral; 44th world congress of vine and wine, 21st General assembly of the OIV, (5 – 9 June 2023), Cádiz, Spain.
- ✓ Physiological responses of red grapevine varieties to heatwave: insights from monitoring physiological activities and drone-based thermal imagery analysis. Oral; 44th world congress of vine and wine, 21st General assembly of the OIV, (5 – 9 June 2023), Cádiz, Spain.
- ✓ Rootstock mediation on grapevine's combined salinity and water deficit stress response: A physiological and metabolic perspective. Oral; 4th Wine Grapevine Symposium, (15/05/2023), Ariel University, Israel.
- ✓ Upshot of salinity and water regime, mediated by rootstock, on cv.Syrah must and wine metabolite profile. Oral: The Gilat and FAAB Arid Agriculture and Biotech Student Symposium, (22/06/2022), Ramat Negev, Israel.

Awards/Achievements

- ✓ Award: Research grant, the International Organisation of Vine and Wine (OIV), France (2023-2026).
- ✓ Award: AKIS Publication Prize at the Ben-Gurion University of the Negev, Israel (2025).
- ✓ Award: Travel grants, Prism Charitable Trust at the Ben-Gurion University of the Negev, Israel (2024).
- ✓ Award: MASOK research grant at the Ben-Gurion University of the Negev, Israel (2024-2025).
- ✓ Award: The Tamar Golan Africa Center at the Ben-Gurion University of the Negev, Israel (2022)
- ✓ Awards: Distinction Honor award for BSc completion, Haramaya University, Ethiopia (2012).
- ✓ Awards and Recognitions: Recognition Silver Medal award in the 7th Ethiopian Science, Technology and Innovation (2016). Award as a team member of the Ethiopian National Sorghum Research Program.

Skills (Technical & Analytical Skills)

Crop Physiology & Plant Phenotyping

- ✓ Gas exchange measurements: LI-COR systems)
- ✓ Leaf chlorophyll fluorescence: Walz MicroPAM
- ✓ Ionomics: ICP-OES ion profiling (Na⁺, Cl⁻, K⁺, Ca²⁺, Mg²⁺, and micro-elements)
- ✓ Root & shoot physiological and biochemical profiling

- ✓ Root phenotyping and root system architecture characterization (RSA)
- ✓ Root architecture trait analysis (root length, angle, density, architecture)

Metabolomics & Molecular Phenotyping

Curriculum Vitae

- ✓ Metabolite identification, spectral curation, annotation, and quantitative analysis (GC-MS/LC-MS targeted and untargeted metabolomics)
- ✓ Volatile profiling (HS-SPME-GC-MS)
- ✓ Transcriptomics integration and analysis
- ✓ Multi-omics integration (mixOmics, R)

Computational Biology, Bioinformatics, Genomics, & Data Science

- ✓ R and Python: statistical modeling, visualization, machine learning, network analysis
- ✓ Predictive modeling, machine learning, and network analysis
- ✓ AI-assisted climate-informed modeling integrating phenotypic and multi-omics datasets
- ✓ Multivariate analysis: PCA, PLS-DA, sPLS-DA, clustering, ANOVA, and more.
- ✓ Metabolic network inference (Gaussian graphical, Bayesian, differential)

Language proficiency

- ✓ Amharic = Native

- ✓ SNP genotyping and GBS analysis (TASSEL)
- ✓ Genomic association studies, trait stability, and performance analysis across environments
- ✓ GxE and predictive modeling, and feature selection

Research & Technical Operations

- ✓ Electronic Lab Notebook (ELN) management and reproducible data workflows
- ✓ Standard Operating Procedures (SOPs), technical reports, and database management
- ✓ Experimental design for multi-environment and controlled environmental crop trials
- ✓ Scientific writing, project coordination, and mentorship
- ✓ Mentoring and training researchers, technicians, and juniors

- ✓ English = Fluent

Memberships & professional services

- ✓ Member: Ethiopian Biotechnology Society, 2021
- ✓ Member: Crop Science Society of Ethiopia, 2016
- ✓ Reviewer: Cogent Food & Agriculture
- ✓ Reviewer: Journal of Crop Improvement
- ✓ Reviewer: African Journal of Plant Science
- ✓ Reviewer: African Journal of Agricultural Research
- ✓ Reviewer: African Journal of Biotechnology
- ✓ Reviewer: Plant physiology and biochemistry
- ✓ Reviewer: Scientia Horticulturae
- ✓ Reviewer: The Plant Cell

References

Up on request